MILITARY SPECIFICATION

PROJECTILE, 105MM, TP-T, M489A1
METAL PARTS ASSEMBLY

This Specification is approved for use within the U.S. Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not covered by the drawings and provides quality assurance provisions for the fabrication of parts, assemblies and packing of projectile 105MM, TP-T, M489A1.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue or the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

TT-C-490	-	Cleaning Methods and Pretreatment of Ferrous Surfaces for Organic Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, Attn: AMSMC-QA, Dover, New Jersey 07801-5001 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-A-2550	-	Ammunition, General Specification For
MIL-A-8625	-	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-A-48078	-	Ammunition, Standard Quality Assurance Provisions, General Specification For

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-171	-	Finishing of Metal and Wood Surfaces
MIL-STD-649	-	Aluminum and Magnesium Products Preparation for Shipment and Storage
MIL-STD-1167	-	Ammunition Data Cards
MIL-STD-1169	-	Packaging, Packing and Marking for shipment for Inert Ammunition Components

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS
(U.S. ARMY ARMAMENT RESEARCH AND DEVELOPMENT CENTER)(ARDC)

7548604	-	Pallet (Boxed Type) For Projectiles and Projectile Metal Parts
9329792	-	Projectile 105MM, TP-T, M489A1 Metal Parts Assembly

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

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American National Standards Institute (ANSI)

ANSI B 46.1 - Surface Texture

(Application for copies should be addressed to: American National Standards Institute (ANSI), 1430 Broadway, New York, NY 10018).

American Society For Testing and Materials (ASTM)

E8	-	Tension Testing of Metallic Materials
B117	-	Salt Spray (Fog) Testing
D785	-	Rockwell Hardness of Plastics and Electrical Insulating Materials
D792	-	Specific Gravity and Density of Plastics by Displacement
D2290	-	Test For Apparent Tensile Strength of Ring or Tubular Plastics by Split Disk Method

(Application for copies should be addressed to: American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA. 19103).

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. Materials, parts, and assemblies shall comply with the requirements specified on the applicable drawings and specifications.

3.2 Heat Treatment or stress relief. Heat treatment or stress relief applied shall be uniform throughout each lot of components and shall be so conducted as to produce in each component the mechanical properties specified on the applicable component drawing.

3.2.1 Cold shaping (if applicable). Cold shaping shall be so accomplished as to produce the required mechanical properties in the metal parts. After cold work, the metal parts shall be stress relieved at the time and temperature specified on the applicable drawing to achieve stress relief and still meet the required mechanical properties.

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3.3 Protective coating. Protective coating shall be in accordance with the applicable drawings and specifications. When protective coating is required by component details, finish requirements will apply to all indicated surfaces of the component regardless of contour configuration. Salt spray testing shall be in accordance with the applicable drawings and specifications.

3.4 Painting. Painting shall comply with the requirements on the applicable drawing. All paint and marking shall be dry to the touch before testing and packing for shipment.

3.5 Aluminum material. Aluminum material used for the body shall be stored under protective covers from adverse conditions as specified on the applicable drawing (see 6.11).

3.6 Proving ground functional tests. The metal parts assemblies shall comply with the following requirements when tested as specified in 4.5.3.

3.6.1 Metal parts security. There shall be no loss, breakup, loosening or deformation of any metal parts component, except the obturator, in the gunbore or in flight, during any of the proving ground tests conducted.

3.6.2 Target-impact-dispersion. The pooled probable error (P.E.) shall not be greater than 0.22 mils vertical and 0.24 mils horizontal when tested in accordance with paragraphs 4.5.3 to 4.5.3.5. See 6.6 for method of calculating pooled P.E.

3.6.3 Flight. There shall be no evidence of erratic flight (see 6.10) greater than specified in Table III paragraph 4.4.3.4.4 and Table IV paragraph 4.4.3.4.5 during any of the proving ground tests conducted.

3.7 First article inspection. This specification contains technical provisions for first article inspection. Requirements for the submission of the first article samples by the contractor shall be as specified in the contract.

3.8 Workmanship. The requirements for workmanship for all components and assemblies are as shown on the applicable component drawings, in the applicable specification and the following:

3.8.1 Metal defects. All metal components shall be free from cracks, splits, cold shuts, inclusions, pipes, bursts, folds, seams, and any other similar defects.

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3.8.2 Molding defects. Soundness of the material shall be in accordance with the applicable drawing and specification. Molded components shall be free of blisters, cracks, check, inclusions, scratches, voids, and any surface or internal defects.

3.8.3 Surface pretreatment. Where cleaning and surface pre-treatment are required by component details, the process shall apply to all indicated surfaces regardless of contour configuration. The protective finish shall be as specified on the applicable drawings and shall be uniformly applied to the surfaces designated by the applicable drawings.

3.8.4 Protective coating. Protective coating shall be uniform and undamaged.

3.8.5 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required by the applicable drawings and shall be in accordance with the provisions of specification MIL-A-2550.

3.8.6 Burrs. No component shall have a burr which might interfere with the assembly or functioning of the projectile or which might be injurious to personnel handling the item.

3.8.7 Foreign matter. All parts and assemblies shall be free of chips, dirt, grease, oil, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. The following types of inspections shall be conducted on this item:

a. First Article Inspection

b. Quality Conformance Inspection

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4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.3. The first article sample shall consist of the items in sample quantities as indicated.

Part Description	Drawings	Quantity
Metal Parts Assembly	9329792	150
Spike	9329794	20
Body	9329793	20
Obturator	9344120	100

One hundred fifty of the above painted metal parts assemblies shall be forwarded to a designated Government facility for loading and subsequent ballistic testing. The sample shall be manufactured using the same materials, equipment, processes, and procedures as will be used in regular production. All parts and materials, including packaging and packing shall be the same as used in regular production and shall be obtained from the same source of supply. A First Article sample or portion thereof, as directed by the procuring agency shall also be submitted whenever there is a lapse in production for a period in excess of 90 days or whenever a change occurs in manufacturing processes, materials used, drawings, specifications or source of supply, such as to significantly affect product uniformity as determined by the Contracting Officer.

4.3.2 Abrasion resistance. After the protective coating has been applied to the bodies of the First Article samples, a specimen taken from the rear end of two different bodies shall be subjected to the abrasion test as specified in MIL-A-8625. If any test specimen (taken from the coated body) fails the requirements of MIL-A-8625, the First Article shall be rejected.

4.3.3 Inspections to be performed. See MIL-A-48078 and Table I specified herein.

4.3.4 Rejection. See MIL-A-48078.

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4.3.5 Process description and controls. The contractor shall prepare a written description of manufacture including process controls for the Body (9329793) and Spike (9329794) as specified in the contract. As a minimum, the process description shall include intermediate steps in press operations and heat treatment/stress relief used to manufacture these parts. Process controls and inspections, including destructive tests used to assure proper quality of those parts and the sample sizes and frequencies shall also be stated. The contractor's written plan of process and process controls shall be prepared in accordance with Data Item Description DI-P-1604. One copy of the plan shall be prepared for each of the following recipients: Commander, ARDC, ATTN: SMCAR-CCH-V and AMSMC-QAT-A(D), Dover, NJ 07801-5001.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS MIL-P-63514A (AR)

PARAGRAPH	TITLE: Projectile, 105MM, TP-T, M489A1, Metal Parts Assembly		SHEET 1 OF 2		DRAWING NUMBER: See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Spike (Dwg. 9329794)				
	Examination for defects (Note 1)	20		3.1	4.4.2.1
	Yield Strength and elongation	5		3.1	4.5
	Body (Dwg. 9329793)				
	Examination for defects (Note 1)	20		3.1	4.4.2.2
	Yield strength and elongation	5		3.1	4.5
	Abrasion test (body specimen)	2		3.1	4.5.2.1
	Protective coating thickness	5		3.1	4.5.2.2
	Obturator (Dwg. 9344120) (Note 2)				
	Examination for defects	100		3.1	4.4.2.3
	Ultimate Load and Extension			3.1	4.5.4
	$140^{o}F \pm 4^{o}F$	20			
	$73^{o}F \pm 4^{o}F$	20			
	$-35^{o}F \pm 4^{o}F$	20			
	Hardness	40		3.1	4.5.5
	Specific Gravity	20		3.1	4.5.6
	Water Absorption and Moisture Content	20		3.1	4.5.7

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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TABLE I. First article inspection.

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63514A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
	Projectile, 105MM, TP-T, M489A1, Metal Parts Assembly				See below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Metal Parts Assembly (Dwg. 9329792)				
	Examination for defects	150		3.1	4.4.2.4
	Ballistic Tests (Notes 3 & 4)	145		3.6	4.5.1
	Salt Spray Test (Note 4)	5		3.3	4.5.2
NOTES:	1. After the examination for defects is completed, these samples can be used for other tests.				
	2. After the dimensional and visual inspection of the 100 samples is completed, subject 60 samples (20 for each temperature) to the ultimate load and extension test. The remaining 40 samples shall be subjected to the hardness test. After the hardness test is completed, subject 20 samples to the specific gravity test and the remaining samples to the water absorption and moisture content test. If the ultimate load and extension test, if 2 or more samples fail to meet the requirements at one temperature phase, or if a total of 3 or more samples fail to meet the requirements at all temperature phases, the First Article shall be rejected.				
	3. See 4.4.3.4 and Table III.				
	4. After the dimensional and visual inspection of the 150 samples is completed, subject 5 samples to the salt spray test in accordance with ASTM-B117. The remaining 145 samples should be subjected to the ballistic test.				
NOTES:					

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. Inspection lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.1.1 Heat treatment lots (if applicable). Heat treatment lots shall contain metal parts forgings from not more than one heat of steel that has been heat treated in the same heat treatment equipment, at the same temperature, in one unchanged continuous process. Lot size shall not exceed 2500 components. The heat treatment lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.2 Stress relieved lots (if applicable). Stress relieved lots shall contain cold work metal parts from not more than one heat of steel that has been cold worked by the same process, utilizing the same tooling and equipment and stress relieved in the same furnace at the same temperature for the same length of time. Lot size shall not exceed 2500 components. The stress relieved lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.3 Stock material lots (if applicable). Stock material lots shall contain machined or forged (if applicable) metal parts from not more than one heat of bar stock steel that has been machined by the same process, tooling and equipment for the same length of time. Lot size shall not exceed 10,000 components. The stock material lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.4 Miscellaneous lots. Small lots (such as metal parts manufactured from abnormally small heats of steel or metal parts that have become separated from previously delivered lots, etc.) may at the discretion of the supplier be grouped into miscellaneous heat treatment, stress relieved or stock material lots as the case may be. Such lots shall be identified by miscellaneous lot code numbers and processed in accordance with 4.4.1.1, 4.4.1.2 or 4.4.1.3 as applicable. The miscellaneous lot thus constituted shall be the basis for the selection of samples for the tension test hereinafter prescribed.

4.4.1.5 Production lots. When it has been determined that bodies of heat treated, stress relieved, stock material or miscellaneous lots have complied with the mechanical properties requirements of the applicable drawing, they may be regrouped

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for assembly into production lots, to which will be assigned a production lot number. The production lot shall consist of projectile assemblies produced by one supplier under one contract in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision, under conditions of continuous production free from interruptions other than those due to the end of a shift, day, or work week. The production lot thus constituted shall be the basis for the selection of samples for functional and protective finish testing hereinafter prescribed.

4.4.2 Examination. See MIL-A-48078.

a. Sampling plan. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

b. Tool control dimensions. Dimensions marked tool control shall be gaged during the First Article testing and once in every 8 hour shift of regular production and prior to any tooling change where destruction of components is necessary to inspect these dimensions. Measurements of the tool may be substituted provided the contractor has established a correlation between the tool dimensions and the component dimensions prior to the start of production. The contractor shall be responsible for maintaining adequate surveillance to assure that drawing requirements are being met throughout the life of the tool. Corrective action will immediately be instituted by the contractor when drawing requirements are not met and all material produced since the last acceptance inspection shall be rejected.

c. Surface texture. The roughness comparison specimens prescribed by ANSI-B 46.1 shall be used as a basis for surface roughness determination.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Spike	1 OF 3	9329794
			NEXT HIGHER ASSEMBLY 9329702

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Pitch diameter of spike thread, min.		0.40%	3.1	Gage
102	Major diameter of spike thread, min.		0.40%	3.1	Gage
103	Spike bourrelet diameter		0.40%	3.1	Gage
104	Runout of front face of spike body shoulder to pitch diameter of spike thread and rear face of spike body shoulder		0.40%	3.1	Gage
105	Diameter of spike nose tip		0.40%	3.1	Gage
106	Runout of spike nose tip to pitch diameter of spike thread and rear face of spike body shoulder		0.40%	3.1	Gage
107	Diameter of spike nose flange		0.40%	3.1	Gage
108	Runout of spike nose flange (outside diameter) to pitch diameter of spike thread and rear face of spike body shoulder		0.40%	3.1	Gage
109	Spike diameter at forward datum		0.40%	3.1	Gage
110	Spike diameter at rear datum		0.40%	3.1	Gage
111	Spike nose tip and nose flange radii, max.		0.40%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Spike	2 OF 3	9329794
			NEXT HIGHER ASSEMBLY 9329792

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
112	Distance from front face of spike nose tip to front face of spike nose flange		0.40%	3.1	Gage
113	Runout of spike nose flange (front face) to pitch diameter of spike thread and rear face of spike body shoulder		0.40%	3.1	Gage
114	Runout of front face of spike nose to pitch diameter of spike thread and rear face of spike body shoulder		0.40%	3.1	Gage
115	Distance from spike nose front face to front spike body shoulder		0.40%	3.1	Gage
116	Runout of spike taper to pitch diameter of spike thread and rear face of spike body shoulder		0.40%	3.1	Gage
117	Front spike body shoulder radius, max.		0.40%	3.1	Gage
118	Tension test	Note 1		3.1	4.5.1
119	Thread damaged or not full		0.40%	3.8.5	Visual
120	Runout of spike thread to bourrelet diameter and rear face of body shoulder		0.40%	3.1	Gage

NOTES:
1. See 4.4.3.1.1 for sample size.
2. Use TT-C-490 to perform this examination.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.1	Spike	3 OF 3	9329794
			NEXT HIGHER ASSEMBLY 9329792

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
MINOR					
201	Distance from rear to rear face of spike body shoulder		0.65%	3.1	Gage
202	Diameter of undercut at rear face of body shoulder		0.65%	3.1	Gage
203	Depth of center drill hole, max., if applicable		0.65%	3.1	Gage
204	Width of spike nose flange		0.65%	3.1	Gage
205	Overall length		0.65%	3.1	Gage
206	Major diameter of spike thread, max.		0.65%	3.1	Gage
207	Length of undercut at the rear face of spike body		0.65%	3.1	Gage
208	Drill hole diameter, max., if applicable		0.65%	3.1	Gage
209	Radii of chamfers, missing or incorrect (except as otherwise classified)		0.65%	3.1	Visual
210	Improper surface pretreatment (Note 2)		0.65%	3.8.3	Visual
211	Metal defective		0.65%	3.8.1	Visual
212	Foreign matter		0 65%	3.8.7	Visual
213	Burr		0.65%	3.8.6	Visual
214	Radii missing or incorrect		0.65%	3.1	Gage
215	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63514A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.2	Body	1 OF 3	9329793
			NEXT HIGHER ASSEMBLY 9329792

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Rear body diameter		0.40%	3.1	Gage
102	Runout of rear body diameter to large diameter of body		0.40%	3.1	Gage
103	Diameter of obturator band seat		0.40%	3.1	Gage
104	Runout of obturator band seat to large diameter of body		0.40%	3.1	Gage
105	Front body diameter		0.40%	3.1	Gage
106	Runout of front body diameter to large diameter of body		0.40%	3.1	Gage
107	Large diameter of body		0.40%	3.1	Gage
108	Pitch diameter of body thread, max.		0.40%	3.1	Gage
109	Minor diameter of body thread, max.		0.40%	3.1	Gage
110	Runout of pitch diameter of body thread to large diameter of body and front body face		0.40%	3.1	Gage
111	Overall body length		0.40%	3.1	Gage
112	Length of obturator band seat		0.40%	3.1	Gage
113	Depth of crimp groove		0.40%	3.1	Gage
114	Pitch diameter of tracer cavity thread, max.		0.40%	3.1	Gage
115	Minor diameter of tracer cavity thread, max.		0.40%	3.1	Gage

NOTES:

DRSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.2	Body		9329793
			NEXT HIGHER ASSEMBLY 9329792

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
116	True position of minor diameter of tracer cavity thread		0.40%	3.1	Gage
117	Diameter of tracer cavity		0.40%	3.1	Gage
118	Length of large body cavity full thread, min.		0.40%	3.1	Gage
119	Length of tracer cavity full thread, min.		0.40%	3.1	Gage
120	Abrasion Test	Note 1		3.1	4.5.2.1.
121	Tension Test	Note 2		3.1	4.5.1.2
122	Runout of rear surface to large diameter of body		0.40%	3.1	Gage
123	Distance from the rear face to the forward edge of large body diameter		0.40%	3.1	Gage
124	Location of crimp groove		0.40%	3.1	Gage
125	Evidence of improper body forging storage after heat treatment, semi-finished body, or finished body		100%	3.5	Visual
126	Thickness of protective finish	Note 3		3.1	4.5.2.2
MINOR					
201	Length of large body cavity		0.65%	3.1	Gage
202	Distance from rear of the body to rear of obturator band seat		0.65%	3.1	Gage

NOTES:
1. See 4.4.3.2.1 for sample size.
2. See 4.4.3.1.2 for sample size.
3. See 4.4.3.2.2 for sample size. Use MIL-STD-171 to perform this inspection.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63514A (AR)

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.2	Body		9329793
			NEXT HIGHER ASSEMBLY 9329792

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
203	True position of tracer cavity to pitch diameter of tracer cavity thread		0.65%	3.1	Gage
204	Depth of tracer cavity		0.65%	3.1	Gage
205	Radius at rear of obturator band seat		0.65%	3.1	Gage
206	Radius at front of obturator band seat		0.65%	3.1	Gage
207	Radius at rear of body		0.65%	3.1	Gage
208	Radius at bottom of tracer cavity		0.65%	3.1	Gage
209	Depth at the front obturator band seat		0.65%	3.1	Gage
210	Depth of tracer cavity thread		0.65%	3.1	Gage
211	Diameter of tracer cavity outer chamfer		0.65%	3.1	Gage
212	Radii of chamfers, missing or incorrect (except as otherwise classified)		0.65%	3.1	Visual
213	Metal defective		0.65%	3.8.1	Visual
214	Foreign matter		0.65%	3.8.7	Visual
215	Burr		0.65%	3.8.6	Visual
216	Radii at the bottom of large body cavity missing or incorrect		0.65%	3.1	Gage
217	Large body cavity diameter at forward datum		0.65%	3.1	Gage
218	Large body cavity diameter at rear datum		0.65%	3.1	Gage
219	Runout of forward face of obturator seat to large body of diameter		0.65%	3.1	Gage
220	Evidence of poor workmanship		0.65%	3.8	Visual
221	Pitch dia. of tracer cavity thread, min.		0.65%	3.1	Gage
222	Pitch dia. of body thread, min.		0.65%	3.1	Gage

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9344120
4.4.2.3	Obturator				NEXT HIGHER ASSEMBLY 9329792
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Inside diameter		0.40%	3.1	Gage
102	Outside diameter		0.40%	3.1	Gage
103	Runout of outside diameter to inside diameter		0.40%	3.1	Gage
104	Total length		0.40%	3.1	Gage
105	Ultimate load, min. a. 140°F ± 4°F b. 73°F ± 4°F c. -35°F ± 4°F	Note 1 & Note 2	2.5%	3.1	4.5.4
106	Extension a. 140°F ± 4°F b. 73°F ± 4°F c. -35°F ± 4°F	Note 1 & Note 2	2.5%	3.1	4.5.4
107	Water absorption, max.	5		3.1	4.5.7
108	Moisture content, max.	3		3.1	4.5.7
109	Hardness, min.	Note 1 & Note 2	2.5%	3.1	4.5.5
110	Specific gravity	3		3.1	4.5.6
111	Arrow marking missing, incorrect or illegible		0.40%	3.1	Visual
112	Diameter of holes		0.40%	3.1	Gage

NOTES:
1. Special inspection level 3 (S-3) and a double sampling plan according to MIL-STD-105D shall be used.
2. Inspection shall be conducted prior to drilling of obturator holes.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63514A(AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.3	Obturator	2 OF 2	9344120
			NEXT HIGHER ASSEMBLY 9329792

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
113	Depth of holes		0.40%	3.1	Gage
114	True position of holes with rear surface and outside diameter		0.40%	3.1	Gage
MINOR					
201	Taper diameter at forward basic datum		0.65%	3.1	Gage
202	Chamfer incorrect		0.65%	3.1	Gage
203	Taper diameter at rear basic datum		0.65%	3.1	Gage
204	Runout of taper diameter at forward basic datum		0.65%	3.1	Gage
205	Runout of taper diameter at rear basic datum		0.65%	3.1	Gage
206	Material defective		0.65%	3.1	Visual
207	Excessive flash		0.65%	3.8.2	Visual
208	Foreign matter		0.65%	3.8.7	Visual
209	Burr		0.65%	3.8.6	Visual
210	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSC N/A FSC 1570, 1 Feb 85

Replaces DESC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 2	DRAWING NUMBER
4.4.2.4	Metal Parts Assembly (After Paint)		9329792
			NEXT HIGHER ASSEMBLY 9343009

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR					
101	Rear bourrelet diameter, max.		0.40%	3.1	Gage
102	Projectile outside diameter, max.		0.40%	3.1	Gage
103	Rear obturator band diameter		0.40%	3.1	Gage
104	Runout of spike nose tip to bourrelet diameter and large diameter of body		0.40%	3.1	Gage
105	Total length, max.		0.40%	3.1	Gage
106	Weight		0.40%	3.1	Approved scale
107	Movement of obturator from application of minimum torque		0.40%	3.1	Gage
108	Gap between spike and body (prior paint)		0.40%	3.1	Gage
109	Improper assembly of obturator		0.40%	3.1	Visual
110	Protective coating inadequate		0.40%	3.3	Visual
111	Salt spray test	Note 1		3.3	4.5.2
112	Ballistic tests	Note 2		3.6	4.5.3
MINOR					
201	Presence of excessive lubricant on obturator seat surface		0.65%	3.1	Visual
202	Presence of paint from rear face of the projectile, including tracer cavity, to front face of obturator groove		0.65%	3.1	Visual

NOTES:
1. See 4.4.3.3 for sample size. Use ASTM-B117 to perform this test.
2. See 4.4.3.4 for sample size.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63514A (AR)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER 9329792
4.4.2.4	Metal Parts Assembly (After Paint)				NEXT HIGHER ASSEMBLY 9343009
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
203	Marking missing, incorrect or illegible		0.65%	3.1	Visual
204	Evidence of poor workmanship		0.65%	3.8	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63514A (AR)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.4.2.5	Carton, Packed (prior to sealing)	1 OF 1	N/A
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR	None defined				
MINOR					
201	Support missing or damaged		0.65%	3.1	Visual
202	Separator missing or damaged		0.65%	3.1	Visual
203	Improper assembly		0.65%	3.1	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63514A (AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Carton, Packed (after sealing)				N/A
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR	None defined				
MINOR					
201	Contents loose		0.65%	3.1	Manual
202	Carton damaged, cut or gouged through all layers		0.65%	3.1	Visual
203	Carton not completely sealed		0.65%	3.1	Visual
204	Marking missing, incorrect or illegible		0.65%	3.1	Visual

NOTES:

AMSMC Form 1570, 1 Feb 85 Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

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4.4.3 Inspection testing. The following tests shall be performed in accordance with the provisions of 4.5.

4.4.3.1 Mechanical properties.

4.4.3.1.1 Spike. Two spikes from each heat treatment, stress relieved, stock material or miscellaneous lot as defined by 4.4.1.1 through 4.4.1.4 as applicable, after final heat treatment or stress relief shall be subjected to this test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any tension test specimen to comply shall be cause for rejection of the lot of spikes represented by the failing specimen.

4.4.3.1.2 Body. Two bodies from each 1200 bodies or fraction thereof shall be subjected to this test to determine compliance with the yield strength and elongation requirements of the applicable drawing. Failure of any specimen to comply shall be cause for rejection of the lot of bodies represented by the failing sample.

4.4.3.2 Protective coating

4.4.3.2.1 Abrasion resistance test (body). Two bodies (a specimen taken from the rear end of each body shall be used) selected from each production lot shall be subjected to the abrasion test as specified in MIL-A-8625. Failure of one or more specimens (taken from the coated body) to comply with the applicable requirements of MIL-A-8625, shall be cause for rejection of the production lot.

4.4.3.2.2 Thickness of protective coating. Two bodies from each production lot shall be subjected to this test to determine compliance with the applicable component drawing. Failure of one or more bodies to comply shall be cause for rejection of the production lot represented by the failing specimen.

4.4.3.3 Salt spray test. Five metal parts assembly samples shall be selected during First Article and submitted for the salt spray test for a duration of 48 hours. Two metal parts assembly samples for every 5000 units or portion thereof shall be selected from each production lot and submitted for the salt spray test for a duration of 48 hours. The acceptance or rejection criteria for the salt spray test is defined in TT-C-490. If any sample fails the salt spray test during First Article inspection, the First Article shall be rejected. If any sample fails the salt spray test during regular production lot acceptance test, the production lot shall be rejected. Production lot salt spray test samples shall comply with paragraph 6.12.

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4.4.3.4 Proving ground functional test. The following proving ground tests will be performed by the government.

4.4.3.4.1 Excess pressure test. The metal part assemblies supplied for the excess pressure phase in accordance with 4.4.3.4.4 and 4.4.3.4.5 will be loaded and assembled into complete rounds. The propellant charge will be adjusted to obtain a chamber pressure of 70,000 psi (482.6 MPa) to 73,900 psi (590.4 MPa).

4.4.3.4.2 Metal parts security. The quantities listed in Table III and IV shall be ballistically tested in the phases indicated. Failure of any projectile assembly to comply with 3.6.1 shall be cause for rejection of the First Article sample or the production lot represented by the sample.

4.4.3.4.3 Target-impact-dispersion. Samples provided in accordance with Table III and IV will be subjected to this test. If the pooled Probable Error (PE) of the sample, calculated in accordance with 6.6, multiplied by the appropriate factor of Table II (see 3.6.2) exceeds the requirements, the lot shall be rejected.

TABLE II

Target impact dispersion factors

Sample Size	Degrees of Freedom	Factor
25 - Test Rounds	20	.80
24 - (one target miss)	19	.80
15 - Calibration Rounds	12	.76
10 - Test or Calibration Rounds	8	.72
9 - (one target miss)	7	.71

4.4.3.4.4 First article sample. The quantities listed in Table III shall be ballistically tested in the phases indicated.

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TABLE III

First article ballistic test

Phase	Temp °F	Tube Wear	Qty	Requirement	Accept	Refer	Reject
A	+ 70°	New	25	a. Security	0	-	1
				b. Flight - Erratic	0	-	1
				c. Target-Impact-Dispersion (Note 2)			
				d. Velocity, Pressure, Spin Rate, for info only			
B	+ 70°	Worn	15	a. Security	0	-	1
				b. Flight - Erratic (Note 1)	1	2	3
				- Target Miss	1	2	-
				c. TID, Velocity, Pressure Spin rate, for info only			
C	Excess Pressure + 70°	New	25	a. Security	0	-	1
				b. Flight - Erratic (Note 1)	1	2	-
				c. TID, Velocity, Pressure (4.4.3.4.1), Spin rate, for info only			
D	+125°	New	25	a. Security	0	-	1
				b. Flight - Erratic (Note 1)	1	2	3
				- Target Miss	2	-	3
				c. TID, Velocity, Pressure, and Spin rate, for info only			

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TABLE III (Continued)

First article ballistic test

Phase	Temp °F	Tube Wear	Qty	Requirement	Accept	Refer	Reject
E	+125°	Worn	15	a. Security	0	-	1
				b. Flight - Erratic (Note 1)	1	2	3
				- Target Miss	2	3	-
				c. TID, Velocity, Pressure and Spin rate for info only			
F	-35°	New	25	a. Security	0	-	1
				b. Flight - Erratic (Note 1)	1	2	3
				- Target Miss	2	-	3
				c. TID, Velocity, Pressure and Spin rate for info only			
G	-35°	Worn	15	a. Security	0	-	1
				b. Flight - Erratic (Note 1)	1	2	3
				- Target Miss	2	-	3
				c. TID, Velocity, Pressure and Spin rate for info only			

NOTE 1. All ballistic phase results, added together with the exception of the 70°F new tube phase, shall reject the first article sample if: three or more erratic flight total occurred, also a combination of five or more erratic flights and targets missed occurred.

2. Target-Impact-Dispersion shall be tested in accordance with 4.4.3.4.3.

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4.4.3.4.5 Production lot sample. The quantities listed in Table IV shall be ballistically tested in the phases indicated.

TABLE IV

Production lot ballistic test

Phase	Tube Wear	Qty	Requirement	Accept	Refer	Reject
+ 70°F	New	15	a. Security	0	-	1
			b. Flight - Erratic	0	-	1
			c. Target-Impact Dispersion (Note 1)			
			d. Velocity, Pressure, Spin rate for first 10 lots for info only			
Excess Pressure +70°F	New	15	a. Security	0	-	1
			b. Flight - Erratic	1	2	-
			c. Velocity, Pressure (4.4.3.4.1) for info only			

Note 1. Target-Impact-Dispersion shall be tested in accordance with 4.4.3.4.3.

4.4.3.4.6 Acceptance criteria. Acceptance criteria for the First Article and production lot samples are shown in TABLE III and IV. Disposition of referred items will be furnished by the Contracting Officer.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein is described in the "Paragraph Reference/Inspection Method" column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.8 herein.

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4.5 Methods of inspection.

4.5.1 Mechanical properties.

4.5.1.1 Spike. From the spikes provided in accordance with 4.4.3.1.1, two tension test specimens conforming to the largest obtainable standard round specimens, as prescribed by ASTM Method E8, shall be taken longitudinally 1 inch from the centerline of the spike to the centerline of the specimen of each sample between the front face of the bourrelet and rear surface of the spike or from any other location deemed necessary by the Government. The criteria for determining yield strength by the offset method shall be as prescribed by ASTM Method E8. The limiting set shall be 0.1 percent.

4.5.1.2 Body. From each sample supplied in accordance with 4.4.3.1.2 for this component, two tension test specimens conforming to the largest obtainable round specimen as prescribed by ASTM Method E8, shall be taken longitudinally from opposite sides of the wall of each sample. The specimen shall be tested using the criteria for performing this test as prescribed by ASTM Method E8. The limiting set shall be 0.2 percent.

4.5.2 Protective coating. Procedures for testing, the cleaning of components, and the application of protective coatings shall be in accordance with the criteria for testing as specified on the applicable component drawings and in the specifications reference thereon. If a salt spray is required on the applicable specification, the salt spray test shall be accomplished on components or assemblies which have been manufactured in accordance with the drawing requirements. The criteria for the salt spray test shall be as prescribed by ASTM-B117.

4.5.2.1 Abrasion resistance. The contractor shall perform abrasion tests on the samples provided by 4.4.3.2.1 in accordance with MIL-A-8625.

4.5.2.2 Thickness of protective coating. The coating thickness on the samples supplied in accordance with 4.4.3.2.2 shall be tested using equipment capable of performing the test accurately. In case of doubt the sample shall be sectionalized and the thickness of coating determined microscopically.

4.5.3 Proving ground functional test.

4.5.3.1 Ballistic phases. The metal parts assemblies shall be subjected to ballistic testing at the phases indicated in Table III (First Article) and Table IV (Production). The metal parts assemblies shall be assembled into complete rounds utilizing the service charge and fired from a fixed mount utilizing a muzzle boresight or an M60 or M1 series tank with an M68 gun.

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4.5.3.1.1 Security. Failure of any projectile assembly to comply with 3.6.1 shall be cause for rejection of the First Article sample or the production lot represented by the sample.

4.5.3.1.2 Target-impact-dispersion. The test and reference samples shall be fired in five round groups (or occasions) with no less than three minutes between rounds and not more than five minutes between rounds. If more than a thirty minute lapse occurs between group firings, or if there was an indication of a trend in the previous group's target impacts to move away from the center of the target, the weapon will be re-zeroed on target center. Target impacts including reference and spotter rounds shall be measured to the nearest centimeter (one half an inch) in both the horizontal and vertical distance from the target center and recorded on the firing record. No more than one lot acceptance test shall be performed on each target. Do not disrupt the firing cadence of a five round group because of velocity equipment failure.

4.5.3.1.2.1 Weapon zeroing - boresight and superelevation. Actual superelevation of the weapon shall be measured and recorded for all target dispersion test phases. This requires that the weapon be zeroed at the beginning, end and during any extended breaks (30 minutes or longer) in target impact test phases using a conventional collimated M26 Pye Watson boresight (or equivalent). Record the weapon elevation and azimuth which aligns the boresight zero on the target center. Then record weapon elevation and azimuth data for all test and calibration phases. The gun superelevation is the weapon elevation during the test phases minus the weapon elevation which produced a muzzle boresight on target center. This data shall be obtained when using either a fixed mount or a service tank weapon system. This information shall be recorded for information only and included in the ballistic test report.

4.5.3.1.2.2 Gun tube backlash removal. After loading and locking a round in the chamber, the gun tube shall be traversed off the target center, dropped down below the target center, traversed to the opposite side of the target center, elevated to the target center elevation, and traversed onto the target center. The distance traversed and dropped should be sufficient to remove the backlash from the system. If the sight is above or beyond the target center, do not reverse direction of gun tube movement, but repeat the sequence.

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4.5.3.1.3 Reference rounds. Reference rounds shall be fired in the target-impact-dispersion phase (70°F - New Tube) in the first article test and in the production lot test. Firing sequence in the first article test shall be: five reference rounds, fifteen test rounds, five reference rounds, ten test rounds, five reference rounds. Firing sequence in the production lot test shall be: five reference rounds, fifteen test rounds, five reference rounds.

4.5.3.1.4 Target. The rounds shall be fired through a target, 20 ft. X 20 ft. minimum (6.10m X 6.10m minimum) placed at 2,000 meters away from the muzzle of the gun when testing with a new tube, and 1,000 meters away from the muzzle of the gun when testing with a worn tube.

4.5.3.1.5 Spotting the target. Spotter rounds shall be fired at the beginning of target-impact-dispersion phase. These rounds shall be temperature conditioned at the same temperature of the representing test phase. Spotter rounds will be used to assure the weapon is zeroed on target center for all test phases indicated. A minimum of three consecutive target hits within one mil of target center is required, and the three hits shall be obtained using the same elevation and deflection weapon settings. After the target has been spotted for a shot group, every effort should be made to keep the elevation and deflection of the weapon the same for that shot group.

4.5.3.1.6 Flight. The observer will be located at an elevated point behind the firing vehicle with an unobstructed view of the line of flight to the target and shall document projectile flight. Any abnormality of flight is to be described (corkscrewing, tumbling, yaw, etc.,) as well as target impact information (hit target, short by 100 meters, left of target, etc.). If a round misses the target with no abnormality of flight, a statement to that effect should also be made.

4.5.3.1.7 Record of firing data. See Tables III and IV, also see paragraphs 4.5.3.1.2.1 and 6.4. All ballistic data results shall be analyzed against the requirements.

4.5.3.2 Temperature tolerances. The following temperature tolerances apply to this specification.

a. -35°F + 5°F (-37°C + 3°C)

b. +70°F ± 2.5°F (+21°C ± 1.5°C)

c. +125°F - 5°F (+52°C - 3°C)

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4.5.3.3 Gun tube gaging. Before each lot is tested, the tubes used shall be star-gaged, both land and groove diameters in both the horizontal and vertical orientations at 25.10 in (637.54mm), 25.25 in (641.35mm), 25.50 in (647.70mm) and 26.00 in (660.40mm) from the rear face of the tube and at one inch (25.4mm) intervals throughout the remaining length of the tube. The star-gage measurements shall be part of the records. See 6.3 for tube wear definitions. The tube shall be cleaned before star-gaged.

4.5.3.4 Photographic equipment. Motion picture or video camera and smear cameras shall be utilized by the Proving Ground to determine compliance with 3.6.1, 3.6.2 and 3.6.3. Motion picture or video camera shall provide clear recording of entire projectile flight path from the gun muzzle to the target. Side smear cameras shall be located at 22 ± 0.5 ft. (6.71 ± .15m) and 105 ± 5 ft. (32.00 ± 1.50m) away from the gun muzzle.

4.5.3.5 Wind velocity. Wind velocity and gusts shall be measured at the firing point prior to and after the target-impact-dispersion test phase, and at the firing point prior to firing each round of the group. The dispersion portion of the ballistic test shall not be fired if the surface wind velocity (nominal) and surface wind gusts exceed 10 knots (18.5 Kilometer per hour (Km/hr)) and 5 knots (9.25 Km/hr) respectively at either equipment location. Only that component of the surface wind normal to the line of fire will be considered. In addition, peak surface wind velocity (nominal plus gusts) normal to the line of fire is not to exceed 10 knots (18.5 Km/hr) during the firing of any group during the dispersion phase.

4.5.3.6 Retest. If for any reason the proving ground considers that test conditions have detrimentally affected the test results, the results shall be invalidated and additional cartridges, as required, shall be tested. A lot shall be retested whenever it fails to meet the target-impact-dispersion requirements as a result of being affected by the following conditions:.

a. The wind velocity exceeds 10 knots (18.5 Km/hr) during the firing of any test group.

b. The cadence between any two consecutive rounds in a test group exceeds 5 minutes.

c. A change was made in the azimuth or elevation of the weapon within a test group.

d. If the test samples and the calibration rounds fired in the target-impact-dispersion phase fail, together, to meet the performance requirements of Section 3 of this specification.

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4.5.3.6.1 Target miss. Production lots which had one target miss during the target-impact-dispersion phase of the proving ground test described herein, shall be retested if the pooled PE of the remaining rounds multiplied by the appropriate factor (see 4.4.3.4.3) meets the requirements of 3.6.2. Two or more target misses by the test rounds shall reject the lot. Target miss coordinates will be estimated and recorded for all test phases, along with pertinent flight observations. Results of retests for target-impact-dispersion shall be referred to the contracting officer for disposition.

4.5.4 Obturator, extension and ultimate load. The ultimate load and extension shall be determined in accordance with ASTM Method D-2290 with the listed exceptions below. The test fixture shall be in accordance with Figure I or its equivalent. Machine loading shall be 0.5 inches per minute ± 0.1 inches per minute. Sample specimens each shall be tested at -35°F ± 4°F (-37°C ± 2°C) 73°F ± 4°F (22°C ± 2°C) and 140°F ± 4° F(60°C ± 2°C). Specimens shall be conditioned for a minimum of two hours prior to performing the test. The ultimate load will be the maximum force recorded during the test and will be independent of cross-sectional area. For elongation the split disc shall be separated until each half is in contact with the specimen without applying a load to the specimen. This distance of separation (initial separation) shall then be measured and recorded. The specimen rupture force shall be measured and recorded. Split disc separation upon specimen rupture (ultimate separation) shall be measured and recorded. Extension shall be determined by subtracting initial separation from ultimate separation. Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot represented by the failing specimen. The specimens used for this test shall be as received obturators. Any test fixture used by the contractor must be approved by the government technical agency. (6.8).

4.5.5 Obturator hardness. Rockwell hardness determination shall be made in accordance with ASTM Method D785. The specimens shall be conditioned for a minimum of twenty-four (24) hours at 73°F ± 4°F (22°C ± 2°C) prior to performing the test. Four (4) readings 90° apart shall be taken on the outside of the sample. Failure of any reading to meet the applicable requirement shall be cause for rejection of the lot represented by the failing specimen.

4.5.6 Obturator specific gravity. Three (3) specimens shall be selected from each of the lots of obturators for use in determining the specific gravity. The test shall be performed in accordance with Method A1 Per ASTM D792. Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot represented by the failing specimen.

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4.5.7 Obturator moisture content and water absorption. The bands shall be wiped clean with an OSHA acceptable solvent, i.e., dichlorotetraflourathane, force air dried, then weighed on an analytical balance. This is the initial weight. The bands are then placed in an oven maintained at 160°F ± 4°F (71°C ± 2°C) for twenty-four (24) hours. Cool in a desiccator and reweigh. Repeat oven conditioning and cooling until constant weight is achieved. This is the conditioned weight. Completely immerse the dried bands in a container of distilled water, maintained at 73°F ± 4°F (22°C ± 2°C) for a period of twenty-four (24) hours minus zero (0) plus one-half (1/2) hour. Bands shall be individually removed from the water, all surface water wiped off with a dry, lint free cloth and weighed immediately on an analytical balance. This gives the wet weight. Percent moisture content shall be calculated as follows:

$$\frac{\text{Initial weight - conditioned weight}}{\text{Initial weight}} \times 100 = \text{\% moisture content}$$

Percent water absorption shall be calculated as follows:

$$\frac{\text{Wet weight - conditioned weight}}{\text{Conditioned weight}} \times 100 = \text{\% water absorption}$$

Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot represented by the failing specimen.

5. PACKAGING

5.1 Preservation and packing.

5.1.1 Level B minimum military pack. The metal parts for Projectile, TP-T, 105MM, M489A1 shall be packed as follows:

5.1.2 Metal parts assembly. Part No. 9329792 shall be packed in accordance with Drawing 7548604 and MIL-STD-1169.

5.1.2.1 Body. Body (Part No. 9329793) after heat treatment and prior to assembly into metal parts assembly (Part No. 9329792) shall be packed in accordance with MIL-STD-649 Level C.

5.1.3 Percentage of extra parts. Packing for the required percentage of extra parts shall be provided for by furnishing the proper number of prime containers.

5.1.4 Shipping. If the components from more than one lot are shipped as a carload, each lot must be kept separate and the division between lots clearly indicated.

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5.1.5 Marking. The pallet shall be marked in accordance with Drawing 7548604, cartons shall be marked in accordance with MIL-STD-1169.

6. NOTES

6.1 Intended use. The projectile metal parts covered by this specification is intended for use with cartridge, 105MM, TP-T, M490A1.

6.2 Ordering data.

6.2.1 Procurement requirements. See MIL-A-48078.

6.2.2 Data requirements. Deliverable data requirements by this specification are cited in the following paragraphs:

Paragraph	Date Requirement	Applicable DID
4.3.5	Process description and controls	DI-P-1604

(Copies of data item descriptions required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

6.2.2.1 Delivery of data to the technical activity. The following address is to be added to those listed in the contract (Form DD 1423) for the delivery of any copy, each, to the technical agency, of the Ammunition Data Cards prepared by the contractor for the duration of the contract: Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAT-A(D), Dover, New Jersey 07801-5001. Ammunition Data Cards shall be prepared in accordance with MIL-STD-1167.

6.3 Gun tube wear limits. For the purpose of this test a new tube is defined as having no greater than .015 inches primary land wear and a midlife tube as having no greater than .030 inches primary land wear. A worn tube is defined as follows:

Primary Land Wear Limits	Secondary Land Wear Limits	Secondary Groove Wear Limits
Greater than 0.030 in. & less than .056 in.	Between 70% & 140% of primary land wear.	Between 80% & 190% of primary land wear.

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6.4 Proving ground test summary. See Tables III and IV.

6.4.1 Firing record information.

6.4.1.1 Target plots and meteorological data. Target impact plots, meteorological data and storage measurements shall also be included in all firing records.

6.4.1.2 Weapon data. The firing records shall include the serial numbers of all weapon and fire control components, velocity reference/calibration ammunition component data where used, and weapon to target azimuth.

6.4.1.3 Velocity - accuracy and tracer function phase. Each round fired during the test shall be identified in the sequence of firing. For each round fired, the record shall contain deflection, elevation, time of firing, muzzle velocity, horizontal and vertical coordinates of impact, and flight characteristics.

6.4.1.4 Analysis of informational data. In the firing tests where velocity, target-impact-dispersion, pressure and spin rate are recorded for information only, the results will be carefully analyzed to determine the quality of the ammunition produced. Evidence of excessive variation will result in an immediate investigation by the contracting officer.

6.5 Visual examination qualification. Standards are to be established at the contractor's plant for compliance with 3.8. When compliance with the applicable requirements is in doubt as a result of visual examination, this characteristic may be measured or gaged to determine acceptability.

6.6 Method of calculating pooled PE. The individual PE results of the groups (5 samples in each group) of samples shall be pooled in accordance with the following formula:

$$PE_{pooled} = \sqrt{\frac{\sum\left[(N_i - 1) \quad PE_i^2\right]}{\sum (N_i - 1)}}$$

Where: i = Number of groups
N_i = Number of samples in a group (target hits)
PE_i = Probable Error of the group

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6.6.1 Probable error calculations. The probable error of the group is calculated in accordance with the following equation:

$$PE_{mils} = \frac{(.6745)\ (1000)\ (SD)}{(X)\ (Rg)}$$

Where: SD - Standard deviation
Rg - Range in meters
X - Conversion factor
(a) when target hits are measured in inches - 39.37.
(b) when target hits are measured in centimeters - 100.0.

6.6.2 Standard deviation. The standard deviation of a sample is calculated in accordance with the following equation found on page 1-10 AMCP 705-110:

$$SD = \sqrt{\frac{N \sum X_i^2 - (\sum X_i)^2}{N\ (N-1)}}$$

Where: N - the number in the sample
X_i - the individual data point

6.6.3 Probable error factor. The factors shown in Table IV are taken from Table A-21 of AMC Pamphlet 706-114 using the 95 percent upper confidence limit (A .05) and $\sum (N_i - 1)$ degrees of freedom. To define $\sum (N_i - 1)$ see 6.6.

6.7 Submission of test data. The following addresses are to be added to the Ballistic Test Request (BTR) for distribution of firing test reports to the technical activity: Commander, US Army Armament Research and Development Center, ATTN: AMSMC-QAT-A(D), SMCAR-CCH-P and SMCAR-CCH-V, Dover, New Jersey 07801-5001.

6.8 Submission of inspection equipment designs for approval. See MIL-A-48078. Submit equipment designs, as required to:

Commander
US Army Research and Development Center
ATTN: AMSMC-QAT-I(D)
Dover, NJ 07801-5001

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6.9 Drawings. Drawings listed in Section 2 of this specification under the heading US Army Armament Research and Development Center (ARDC) may also include drawings prepared by, and identified as Edgewood Arsenal, Frankford Arsenal, Rock Island Arsenal ARRADCOM or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under the cognizance of ARDC.

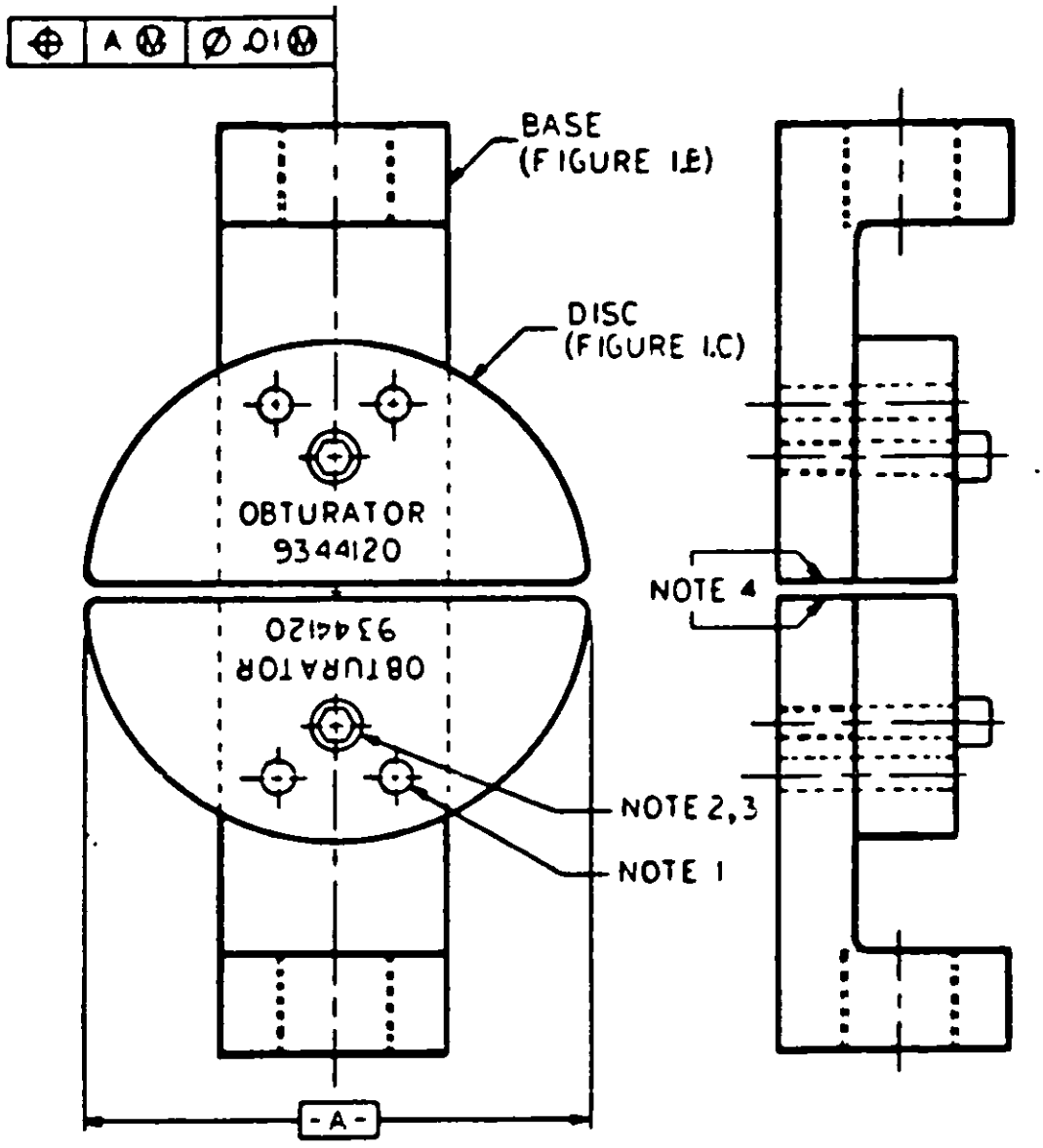
6.10 Erratic flight definition. Erratic flight is any abnormality in the projectile flight path, usually displayed by corkscrewing, tumbling or undulation.

6.11 Adverse conditions. The material used for the body is sensitive to stress corrosion after heat treatment. Stress corrosion is aggravated by exposure to the natural elements especially moisture in any form, i.e., rain, snow, sleet, dew, condensation, etc. Suitable protective covers following heat treatment of the body is required.

6.12 Salt spray test specimens. Salt spray samples shall be components/assemblies randomly selected from each inspection lot. The contractor may elect to use reject components/assemblies for salt spray samples. If the contractor elects to use rejected components/assemblies, a request shall be forwarded to the contracting officer and approval received prior to use. A detailed plan describing the selection and processing of the sample shall accompany the request and shall serve as the basis for approval. The plan must clearly establish the relationship of the test specimens to the overall manufacturing process and lotting criteria which will apply to the components being accepted. Specifically, the plan shall address how test specimens are selected and identified; how they are processed at the same time and in the same way as the lot they represent; and how they are randomly introduced and distributed when the protective finish is applied. In addition, procedures shall be included for the contractor to require his quality control staff together with the resident DCAS quality assurance representative, to maintain surveillance over the process to ensure test specimens are not lost or given preferential treatment.

6.13 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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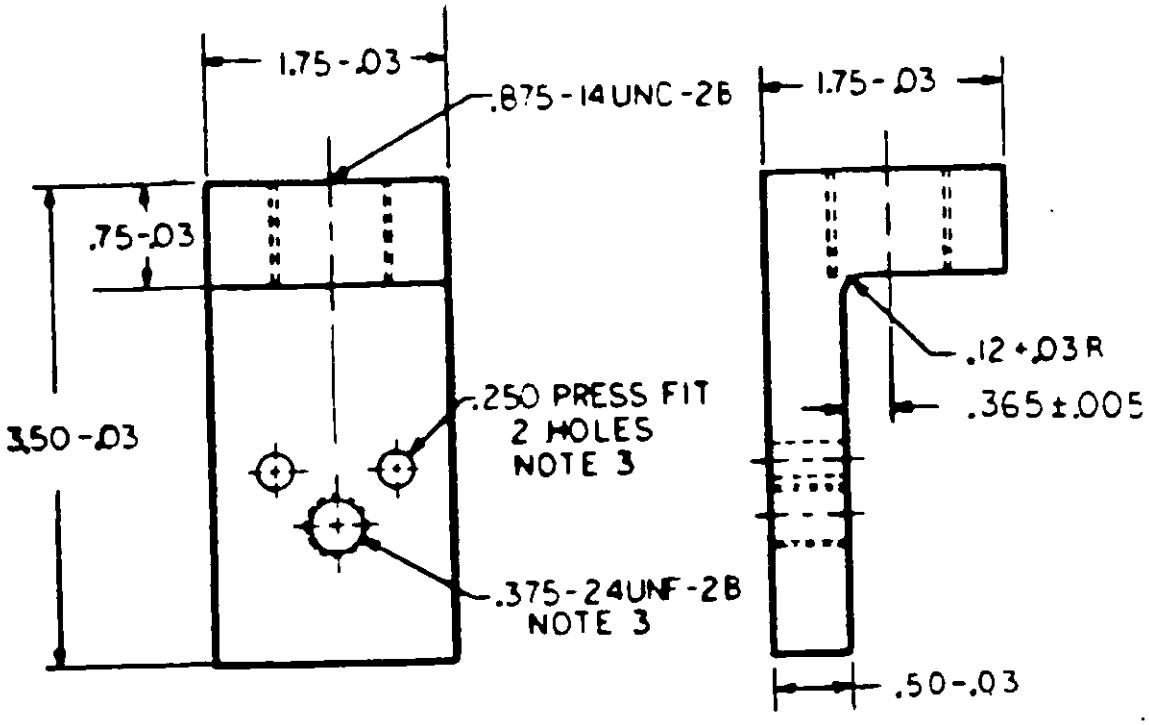


NOTES:-

1. STEEL DOWEL PIN, .250-.005 DIAMETER, 1 3/16 LONG, 4 REQUIRED.
2. SOCKET HEAD CAP SCREW, .375-24UNF-2A, 1 3/16 LONG, 2 REQUIRED.
3. ASSEMBLE SOCKET HEAD CAP SCREW WITH 50 INCH POUND TORQUE MINIMUM.
4. BASE MUST NOT PROTRUDE PAST DISC.

FIGURE 1.A DISC AND BASE ASSEMBLY (OBTURATOR)

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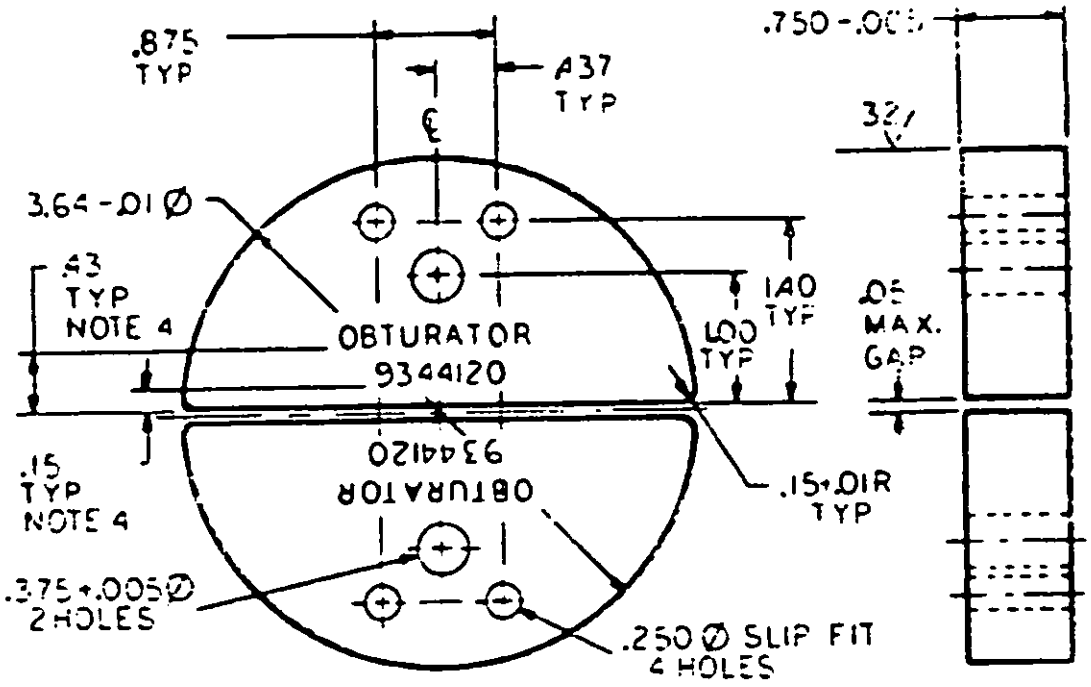


NOTES:-

1. MATERIAL:- STEEL, HOT-ROLLED ALLOY BARS, GRADE 4340, ASTM-A322.
2. FINISH ALL OVER 125.
3. LOCATE HOLES FROM DISC HOLE PATTERN (FIGURE 1.C).
4. BASE, 2 REQUIRED.

FIGURE 1.B BASE (OBTURATOR)

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NOTES.-

1. MATERIAL:- STEEL, HOT-ROLLED ALLOY BARS, GRADE 4340, ASTM-A322.
2. FINISH ALL OVER 63, EXCEPT AS NOTED.
3. DISC (2 REQUIRED) MAY BE FABRICATED FROM ONE PIECE BY SLITTING. MAXIMUM GAP .05 PERMITTED.
4. METAL STAMP AS SHOWN WITH APPROXIMATELY .125 HIGH LETTERS AND NUMBERS.
5. Tolerances: ±.01 (2 place dimensions), ±.010 (3 place dimensions). unless otherwise specified.

FIGURE 1.C DISC (OBTURATOR)

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Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1315-A778)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-63514A	PROJECTILE, 105MM, TP-T, M489A! METAL PARTS ASSEMBLY

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)

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